

Các phân phối thông dụng

Đều rời rạc

$$P(X = x_i) = \frac{1}{N}, \quad i = 1, \ldots, N,$$

$$\operatorname{E}(X) = \frac{1}{N} \sum_{i=1}^N x_i,$$

$$\operatorname{Var}(X) = \frac{1}{N} \sum_{i=1}^N [x_i - \operatorname{E}(X)]^2.$$

Có hoàn lại: nhĩ thức *B*(*n*, *p*), *p* = *K*/*N*

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n - k}, \quad k = 0, 1, \ldots, n,$$

$$\operatorname{E}(X) = np, \quad \operatorname{Var}(X) = np(1 - p).$$

*B*(*n*, *p*) ≈ *N*(*np*, *np*(1 − *p*)) (*n* ≥ 50, *np* ≥ 5, *n*(1 − *p*) ≥ 5).

Không hoàn lại: siêu bội *HG*(*N*, *K*, *n*), *p* = *K*/*N*

$$P(X = k) = \frac{\binom{K}{k} \binom{N - K}{n - k}}{\binom{N}{n}},$$

$$\operatorname{E}(X) = np, \quad \operatorname{Var}(X) = np(1 - p) \frac{N - n}{N - 1}.$$

Poisson *P*(*μ*)

$$P(X = k) = e^{-\mu} \frac{\mu^k}{k!}, \quad k = 0, 1, \ldots; \quad \operatorname{E}(X) = \mu, \quad \operatorname{Var}(X) = \mu.$$

Chuẩn *N*(*μ*, *σ*<sup>2</sup>)

$$f_X(x) = \frac{1}{\sigma \sqrt{2\pi}} \exp \left[ -\frac{(x - \mu)^2}{2\sigma^2} \right], \quad x \in \mathbb{R},$$

$$\operatorname{E}(X) = \mu, \quad \operatorname{Var}(X) = \sigma^2.$$

Chi bình phương *χ*<sup>2</sup>(*n*)

$$f_Z(z) = \frac{z^{n/2 - 1} e^{-z/2}}{2^{n/2} \Gamma(n/2)}, \quad z \geq 0,$$

$$\operatorname{E}(Z) = n, \quad \operatorname{Var}(Z) = 2n.$$

Student *St*(*n*)

$$f_Z(z) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{n\pi} \, \Gamma(\frac{n}{2})} \left( 1 + \frac{z^2}{n} \right)^{-\frac{n+1}{2}}, \quad z \in \mathbb{R};$$

$$\operatorname{E}(Z) = 0 \; (n > 1), \quad \operatorname{Var}(Z) = \frac{n}{n - 2} \; (n > 2).$$

Fisher *F*(*m*, *n*)

$$f_Z(z) = \frac{\Gamma(\frac{m+n}{2}) \left(\frac{m}{n}\right)^{m/2} z^{m/2 - 1}}{\Gamma(\frac{m}{2}) \Gamma(\frac{n}{2}) \left(1 + \frac{m}{n} z\right)^{(m+n)/2}}, \quad z > 0;$$

$$\operatorname{E}(Z) = \frac{n}{n - 2} \; (n > 2), \quad \operatorname{Var}(Z) = \frac{2n^2(m + n - 2)}{m(n - 2)^2(n - 4)} \; (n > 4).$$

Đa thức

$$P(X_1 = x_1, \ldots, X_k = x_k) = \frac{n!}{x_1! \cdots x_k!} p_1^{x_1} \cdots p_k^{x_k},$$

$$\sum_{i=1}^k x_i = n, \quad p_k = 1 - \sum_{i=1}^{k-1} p_i,$$

$$\operatorname{E}(X_i) = np_i, \quad \operatorname{Var}(X_i) = np_i(1 - p_i).$$

Phân phối mẫu của trung bình và phương sai

$$\overline{X} \perp S^2, \quad \overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \quad \frac{(n - 1)S^2}{\sigma^2} \sim \chi^2(n - 1),$$

$$\frac{\overline{X} - \mu}{S/\sqrt{n}} \sim T(n - 1).$$

$$Y_n = \sqrt{n} \frac{\overline{X} - \mu}{\sigma} = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \approx N(0, 1).$$

$$n \geq 30: \quad \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \approx N(0, 1), \quad \frac{\overline{X} - \mu}{S/\sqrt{n}} \approx N(0, 1).$$

1. Ước lượng và tính chất ước lượng

**Ước lượng không chệch:** *E*(*θ̂*) = *θ*. **Độ chệch** (**Bias**): *B*(*θ̂*) = *E*(*θ̂*) − *θ*.

$$\operatorname{MSE}(\hat{\theta}) = \operatorname{E}[(\hat{\theta} - \theta)^2] = \operatorname{Var}(\hat{\theta}) + (\operatorname{Bias}(\hat{\theta}))^2.$$

*θ̂*<sub>1</sub> hiệu quả hơn *θ̂*<sub>2</sub> nếu cả hai đều không chệch và *Var*(*θ̂*<sub>1</sub>) < *Var*(*θ̂*<sub>2</sub>).

- Cho trước hai ước lượng, *θ̂* và *θ̂*, tiêu chuẩn MSE cho phép ta chọn *θ̂* nếu, với cỡ mẫu *n*,

$$\operatorname{MSE}(\hat{\theta}) < \operatorname{MSE}(\hat{\tilde{\theta}}),$$

- hoặc

$$\operatorname{Var}(\hat{\theta}) - \operatorname{Var}(\hat{\tilde{\theta}}) > (\operatorname{Bias}(\hat{\theta}))^2 - (\operatorname{Bias}(\hat{\tilde{\theta}}))^2.$$

**Ước lượng vững:**

$$\lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| > \epsilon) = 0, \quad \forall \epsilon > 0.$$

**Mômen tổng thể và mômen mẫu:**

$$\mu_j'(\theta) = \operatorname{E}(X^j) = \int x^j f(x) \, dx, \quad m_j = \frac{1}{n} \sum_{i=1}^n X_i^j.$$

**Định lý Chebyshev:**

$$P(|X - \operatorname{E}(X)| \geq \epsilon) \leq \frac{\operatorname{Var}(X)}{\epsilon^2}.$$

**Hàm hợp lý:**

$$L(\theta) = L(\theta \mid X) = L(\theta_1, \ldots, \theta_n \mid X_1, \ldots, X_n)$$

$$= \prod_{i=1}^n f(X_i \mid \theta_1, \ldots, \theta_n).$$

2. Khoảng tin cậy

Sai số ước lượng:

$$\epsilon = \begin{cases} z_{1-\alpha/2} \frac{\sigma}{\sqrt{\frac{n}{s}}}, \\ z_{1-\alpha/2} \frac{\sigma}{\sqrt{\frac{n}{s}}}, \\ t_{1-\alpha/2}^{n-1} \frac{\sigma}{\sqrt{\frac{n}{s}}}, \end{cases} \quad \epsilon = z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}.$$

3. Kiểm định giả thuyết

**Hàm năng lực:**

$$\beta(\theta) = P(\text{Bác bỏ } H_0 \mid \theta) = P(\text{Thông kê kiểm định} \in C \mid \theta).$$

**Năng lực tại *θ*<sub>1</sub>:** Giá trị hàm năng lực tại điểm cụ thể *θ*<sub>1</sub> ∈ *H*<sub>1</sub>, ký hiệu *β*(*θ*<sub>1</sub>) = 1 − *β* (*β* là xác suất sai lầm loại II).

**Mức ý nghĩa** (*α*):

$$\alpha = \max_{\theta \in H_0} \beta(\theta) = P(\text{Bác bỏ } H_0 \mid H_0 \text{ đúng}).$$

**Kỹ vng, trường hợp biết *σ*<sup>2</sup>**

$$Z_0 = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}.$$

| Giả thuyết                                                              | Miền bác bỏ                                                                                           |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <i>H</i> <sub>0</sub> <span> </span> : <i>μ</i> = <i>μ</i> <sub>0</sub> | <i>C</i> = { <i>x</i> <span> </span> :   <i>z</i> <sub>0</sub>   > <i>z</i> <sub>1−<i>α</i>/2</sub> } |
| <i>H</i> <sub>1</sub> <span> </span> : <i>μ</i> ≠ <i>μ</i> <sub>0</sub> |                                                                                                       |
| <i>H</i> <sub>0</sub> <span> </span> : <i>μ</i> = <i>μ</i> <sub>0</sub> | <i>C</i> = { <i>x</i> <span> </span> : <i>z</i> <sub>0</sub> < − <i>z</i> <sub>1−<i>α</i></sub> }     |
| <i>H</i> <sub>1</sub> <span> </span> : <i>μ</i> < <i>μ</i> <sub>0</sub> |                                                                                                       |
| <i>H</i> <sub>0</sub> <span> </span> : <i>μ</i> = <i>μ</i> <sub>0</sub> | <i>C</i> = { <i>x</i> <span> </span> : <i>z</i> <sub>0</sub> > <i>z</i> <sub>1−<i>α</i></sub> }       |
| <i>H</i> <sub>1</sub> <span> </span> : <i>μ</i> > <i>μ</i> <sub>0</sub> |                                                                                                       |

**Kỹ vng, trường hợp không biết *σ*<sup>2</sup>, mẫu nhỏ**

$$T_0 = \frac{\overline{X} - \mu_0}{S/\sqrt{n}}.$$

| Miền bác bỏ                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------|
| <i>C</i> = { <i>x</i> <span> </span> :   <i>t</i> <sub>0</sub>   > <i>t</i> <sub>1−<i>α</i>/2</sub> ( <i>n</i> − 1)} |
| <i>C</i> = { <i>x</i> <span> </span> : <i>t</i> <sub>0</sub> < − <i>t</i> <sub>1−<i>α</i></sub> ( <i>n</i> − 1)}     |
| <i>C</i> = { <i>x</i> <span> </span> : <i>t</i> <sub>0</sub> > <i>t</i> <sub>1−<i>α</i></sub> ( <i>n</i> − 1)}       |

Tỷ lệ

$$Z_0 = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)}/n}.$$

| Giả thuyết                                                              | Miền bác bỏ                                                                                           |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <i>H</i> <sub>0</sub> <span> </span> : <i>p</i> = <i>p</i> <sub>0</sub> | <i>C</i> = { <i>x</i> <span> </span> :   <i>z</i> <sub>0</sub>   > <i>z</i> <sub>1−<i>α</i>/2</sub> } |
| <i>H</i> <sub>1</sub> <span> </span> : <i>p</i> ≠ <i>p</i> <sub>0</sub> |                                                                                                       |
| <i>H</i> <sub>0</sub> <span> </span> : <i>p</i> = <i>p</i> <sub>0</sub> | <i>C</i> = { <i>x</i> <span> </span> : <i>z</i> <sub>0</sub> < − <i>z</i> <sub>1−<i>α</i></sub> }     |
| <i>H</i> <sub>1</sub> <span> </span> : <i>p</i> < <i>p</i> <sub>0</sub> |                                                                                                       |
| <i>H</i> <sub>0</sub> <span> </span> : <i>p</i> = <i>p</i> <sub>0</sub> | <i>C</i> = { <i>x</i> <span> </span> : <i>z</i> <sub>0</sub> > <i>z</i> <sub>1−<i>α</i></sub> }       |
| <i>H</i> <sub>1</sub> <span> </span> : <i>p</i> > <i>p</i> <sub>0</sub> |                                                                                                       |

4. So sánh hai kỹ vng

**Biết phương sai**

$$Z_0 = \frac{\overline{X} - \overline{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}}.$$

| Đối thuyết                                                                                                   | Miền bác bỏ                                                |
|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <i>H</i> <sub>1</sub> <span> </span> : <i>μ</i> <sub>1</sub> − <i>μ</i> <sub>2</sub> ≠ <i>D</i> <sub>0</sub> | <i>z</i> <sub>0</sub>   > <i>z</i> <sub>1−<i>α</i>/2</sub> |
| <i>H</i> <sub>1</sub> <span> </span> : <i>μ</i> <sub>1</sub> − <i>μ</i> <sub>2</sub> < <i>D</i> <sub>0</sub> | <i>z</i> <sub>0</sub> < − <i>z</i> <sub>1−<i>α</i></sub>   |
| <i>H</i> <sub>1</sub> <span> </span> : <i>μ</i> <sub>1</sub> − <i>μ</i> <sub>2</sub> > <i>D</i> <sub>0</sub> | <i>z</i> <sub>0</sub> > <i>z</i> <sub>1−<i>α</i></sub>     |

$$\sigma_1^2 = \sigma_2^2 = \sigma^2$$

$$S_p^2 = \frac{(n - 1)S_1^2 + (m - 1)S_2^2}{n + m - 2}, \quad T_0 = \frac{\overline{X} - \overline{Y} - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n} + \frac{1}{m}}}.$$

| Đối thuyết                                                                                                   | Miền bác bỏ                                                                     |
|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <i>H</i> <sub>1</sub> <span> </span> : <i>μ</i> <sub>1</sub> − <i>μ</i> <sub>2</sub> ≠ <i>D</i> <sub>0</sub> | <i>t</i> <sub>0</sub>   > <i>t</i> <sub>1−<i>α</i>/2</sub> <sup><i>df</i></sup> |
| <i>H</i> <sub>1</sub> <span> </span> : <i>μ</i> <sub>1</sub> − <i>μ</i> <sub>2</sub> < <i>D</i> <sub>0</sub> | <i>t</i> <sub>0</sub> < − <i>t</i> <sub>1−<i>α</i></sub> <sup><i>df</i></sup>   |
| <i>H</i> <sub>1</sub> <span> </span> : <i>μ</i> <sub>1</sub> − <i>μ</i> <sub>2</sub> > <i>D</i> <sub>0</sub> | <i>t</i> <sub>0</sub> > <i>t</i> <sub>1−<i>α</i></sub> <sup><i>df</i></sup>     |

$$\sigma_1^2 \neq \sigma_2^2$$

$$T_0 = \frac{\overline{X} - \overline{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n} + \frac{S_2^2}{m}}},$$

$$df = \frac{\left[ \left(\frac{s_1^2}{n}\right) + \left(\frac{s_2^2}{m}\right) \right]^2}{\left(\frac{s_1^2}{n}\right)^2 + \left(\frac{s_2^2}{m}\right)^2}.$$

$$\frac{\left(\frac{s_1^2}{n}\right)}{n - 1} + \frac{\left(\frac{s_2^2}{m}\right)}{m - 1}$$

Trạ bảng tương tự trường hợp các *σ*<sup>2</sup> bằng nhau.

5. Hai mẫu phụ thuộc (nguồn: 2 mẫu không độc lập)

$$D_i = X_{1i} - X_{2i}, \quad \bar{d} = \frac{1}{n} \sum_{i=1}^n d_i,$$

$$s_{\bar{d}}^2 = \frac{1}{n - 1} \sum_{i=1}^n (d_i - \bar{d})^2 = \frac{1}{n - 1} \sum_{i=1}^n d_i^2 - \frac{n}{n - 1} (\bar{d})^2,$$

$$T_0 = \frac{\overline{D} - D_0}{S_D/\sqrt{n}}.$$

| Đối thuyết                                                                                  | Miền bác bỏ                                                                      |
|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <i>H</i> <sub>1</sub> <span> </span> : <i>μ</i> <sub><i>D</i></sub> ≠ <i>D</i> <sub>0</sub> | <i>t</i> <sub>0</sub>   > <i>t</i> <sub>1−<i>α</i>/2</sub> <sup><i>n</i>−1</sup> |
| <i>H</i> <sub>1</sub> <span> </span> : <i>μ</i> <sub><i>D</i></sub> < <i>D</i> <sub>0</sub> | <i>t</i> <sub>0</sub> < − <i>t</i> <sub>1−<i>α</i></sub> <sup><i>n</i>−1</sup>   |
| <i>H</i> <sub>1</sub> <span> </span> : <i>μ</i> <sub><i>D</i></sub> > <i>D</i> <sub>0</sub> | <i>t</i> <sub>0</sub> > <i>t</i> <sub>1−<i>α</i></sub> <sup><i>n</i>−1</sup>     |

6. So sánh hai tỷ lệ

$$Z_0 = \frac{\hat{p}_1 - \hat{p}_2 - D_0}{\sqrt{\hat{p}(1 - \hat{p}) \left( \frac{1}{n} + \frac{1}{m} \right)}},$$

$$\hat{p}_1 = \frac{X}{n}; \quad \hat{p}_2 = \frac{Y}{m}; \quad \hat{p} = \frac{X + Y}{n + m}.$$

| Đối thuyết                                                                                                   | Miền bác bỏ                                                |
|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <i>H</i> <sub>1</sub> <span> </span> : <i>p</i> <sub>1</sub> − <i>p</i> <sub>2</sub> ≠ <i>D</i> <sub>0</sub> | <i>z</i> <sub>0</sub>   > <i>z</i> <sub>1−<i>α</i>/2</sub> |
| <i>H</i> <sub>1</sub> <span> </span> : <i>p</i> <sub>1</sub> − <i>p</i> <sub>2</sub> < <i>D</i> <sub>0</sub> | <i>z</i> <sub>0</sub> < − <i>z</i> <sub>1−<i>α</i></sub>   |
| <i>H</i> <sub>1</sub> <span> </span> : <i>p</i> <sub>1</sub> − <i>p</i> <sub>2</sub> > <i>D</i> <sub>0</sub> | <i>z</i> <sub>0</sub> > <i>z</i> <sub>1−<i>α</i></sub>     |

7. Hồi quy tuyến tính

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2,$$

$$S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n y_i^2 - n\bar{y}^2,$$

$$S_{xy} = S_{yx} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}.$$

$$SST = S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2,$$

$$SSR = \beta_1^2 S_{xx} = \frac{S_{xy}^2}{S_{xx}},$$

$$SSE = SST - SSR = \sum_{i=1}^n (y_i - \hat{y}_i)^2,$$

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST},$$

$$MSE = \sigma_{est}^2 = \frac{SSE}{n - 2}.$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

**Khoảng tin cậy**

$$\left( \hat{\beta}_1 - t_{1-\alpha/2}^{n-2} \sqrt{\frac{MSE}{S_{xx}}}; \; \hat{\beta}_1 + t_{1-\alpha/2}^{n-2} \sqrt{\frac{MSE}{S_{xx}}} \right),$$

$$\left( \hat{\beta}_0 - t_{1-\alpha/2}^{n-2} \sqrt{MSE \left( \frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)}; \; \hat{\beta}_0 + t_{1-\alpha/2}^{n-2} \sqrt{MSE \left( \frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)} \right).$$

**Kiểm định *β*<sub>0</sub>**

$$t_{\beta_0} = \frac{\hat{\beta}_0 - b_0}{\sqrt{MSE \left( \frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)}}.$$

| Đối thuyết                                                                           | Miền bác bỏ                                                                                         |
|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <i>H</i> <sub>1</sub> <span> </span> : <i>β</i> <sub>0</sub> ≠ <i>b</i> <sub>0</sub> | <i>t</i> <sub><i>β</i><sub>0</sub></sub>   > <i>t</i> <sub>1−<i>α</i>/2</sub> <sup><i>n</i>−2</sup> |
| <i>H</i> <sub>1</sub> <span> </span> : <i>β</i> <sub>0</sub> < <i>b</i> <sub>0</sub> | <i>t</i> <sub><i>β</i><sub>0</sub></sub> < − <i>t</i> <sub>1−<i>α</i></sub> <sup><i>n</i>−2</sup>   |
| <i>H</i> <sub>1</sub> <span> </span> : <i>β</i> <sub>0</sub> > <i>b</i> <sub>0</sub> | <i>t</i> <sub><i>β</i><sub>0</sub></sub> > <i>t</i> <sub>1−<i>α</i></sub> <sup><i>n</i>−2</sup>     |

**Kiểm định *β*<sub>1</sub>**

$$t_{\beta_1} = \frac{\hat{\beta}_1 - b_1}{\sqrt{MSE/S_{xx}}}.$$

| Đối thuyết                                                                           | Miền bác bỏ                                                                                         |
|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <i>H</i> <sub>1</sub> <span> </span> : <i>β</i> <sub>1</sub> ≠ <i>b</i> <sub>1</sub> | <i>t</i> <sub><i>β</i><sub>1</sub></sub>   > <i>t</i> <sub>1−<i>α</i>/2</sub> <sup><i>n</i>−2</sup> |
| <i>H</i> <sub>1</sub> <span> </span> : <i>β</i> <sub>1</sub> < <i>b</i> <sub>1</sub> | <i>t</i> <sub><i>β</i><sub>1</sub></sub> < − <i>t</i> <sub>1−<i>α</i></sub> <sup><i>n</i>−2</sup>   |
| <i>H</i> <sub>1</sub> <span> </span> : <i>β</i> <sub>1</sub> > <i>b</i> <sub>1</sub> | <i>t</i> <sub><i>β</i><sub>1</sub></sub> > <i>t</i> <sub>1−<i>α</i></sub> <sup><i>n</i>−2</sup>     |

8. Hiệp phương sai và tương quan

$$\begin{aligned} \sigma_{XY} &= \operatorname{Cov}(X, Y) \\ &= \operatorname{E}[(X - \operatorname{E}(X))(Y - \operatorname{E}(Y))] \\ &= \operatorname{E}(XY) - \operatorname{E}(X)\operatorname{E}(Y). \end{aligned}$$

$$\rho_{XY} = \frac{\operatorname{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\sigma_{XY}}{\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}}.$$

$$r_{xy} = \frac{s_{xy}}{s_x s_y} = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}.$$